INTERMEDIATE PRACTICE WORKSHEET

Chapter 13: Fundamentals of Algebraic Expressions & Polynomials

Grade 7 / Class VII Mathematics — **Time Allowed:** 60 Minutes — **Max Marks:** 50

Instructions:

- Read every question carefully before answering.
- Show neat step-by-step working for all subjective questions.
- Complete **Part I (Questions)** first before checking your answers against **Part II (Detailed Solutions)** at the end of the document.

PART I: PRACTICE QUESTIONS (50 Marks)

SECTION A: Multiple Choice Questions (6 Marks)

Select the single correct option for each question. (1 Mark each)

Q1. What is the degree of the multi-variable polynomial $5x^3y^2 - 7x^2y^4 + 3xy^3 - 11$?

(a) 5 (b) 6 (c) 4 (d) 7

Q2. The verbal statement *"Five times the square of a subtracted from the product of* $3b$ *and the sum of* x *and* y*"* translates to:

(a) $5a^2 - 3b(x+y)$
(b) $3b(x+y) - 5a^2$
(c) $(3b + x + y) - 5a^2$
(d) $3bxy - 25a^2$

Q3. In the algebraic term $-\frac{4}{9}p^3q^2r$, the coefficient of p^2qr is:

(a) $-\frac{4}{9}p$ (b) $-\frac{4}{9}pq$ (c) $\frac{4}{9}p^2$ (d) $-\frac{4}{9}p^2q$

Q4. Which of the following expressions is a valid polynomial?

(a) $3x^2 - 5\sqrt{x} + 2$
(b) $\frac{2}{3}x^4 - \frac{7}{x^2} + 9$
(c) $\frac{\sqrt{5}}{2}x^3 - \frac{3}{4}x^2 + \sqrt{2}x - 1$
(d) $x^{3/2} + 4x - 8$

Q5. Subtracting $(-3x^2 + 5xy - 2y^2)$ from $(2x^2 - 3xy + 4y^2)$ yields:

(a) $-x^2 + 2xy + 2y^2$
(b) $5x^2 - 8xy + 6y^2$
(c) $5x^2 + 2xy - 6y^2$
(d) $-5x^2 + 8xy - 6y^2$

Q6. If $x = -2$, $y = 3$, and $z = -\frac{1}{2}$, what is the value of $4x^2y - 2xyz + 8z^2$?

(a) 38 (b) 46 (c) 34 (d) 50

SECTION B: Short Answer & Conceptual Questions (10 Marks)

Q7. **[2 Marks]** State whether each of the following expressions is a polynomial. If it is a polynomial, write its degree and classify it as a monomial, binomial, or trinomial:

(i) $E_1(x, y) = \frac{3}{5}x^2y^3 - 7x^4y + 2$

(ii) $E_2(a) = 4a^2 - \frac{3}{a} + 5$

Q8. [**2 Marks**] Find the specific coefficient requested:

(i) Find the coefficient of x^2y in the term $-15x^3y^2z$.

(ii) Find the coefficient of $-4ab$ in the term $12a^2b^3c$.

Q9. [**3 Marks**] Simplify the expression by grouping and combining like terms:

$$3x^3 - 5x^2y + 2xy^2 - 7x^3 + 8x^2y - 6xy^2 + 4x^3 - 3x^2y$$

Q10. [**3 Marks**] What polynomial must be added to $(2a^2 - 5ab + 3b^2)$ so that the resulting sum becomes $(a^2 + 2ab - b^2)$?

SECTION C: Expression Formulation & Operations (12 Marks)

Q11. [**3 Marks**] Formulate an algebraic expression for each of the following verbal descriptions:

(i) The sum of the squares of p and q divided by thrice their product.

(ii) Seven times m diminished by the quotient when n is divided by 5.

(iii) The excess of $(5x^2 - 2xy + 3y^2)$ over $(2x^2 + 3xy - 4y^2)$.

Q12. [**3 Marks**] Subtract the sum of $(3x^2 - 5x + 7)$ and $(-2x^2 + 8x - 11)$ from the sum of $(5x^2 - 2x + 1)$ and $(x^2 - 3x - 4)$.

Q13. [**3 Marks**] Given three polynomials:

$$A = 2x^2 - 3xy + y^2, \quad B = -x^2 + 2xy + 3y^2, \quad C = 3x^2 - xy - 2y^2$$

Evaluate the simplified expression for:

(i) $A + B - C$

(ii) $2A - 3B + C$

Q14. [**3 Marks**] The perimeter of a rectangle is represented by $(14x^2 + 6x - 8)$ units. If the length of one side is $(4x^2 - x + 3)$ units, find the algebraic expression for the adjacent side (breadth).

SECTION D: Advanced Application Problems (22 Marks)

Q15. [**5 Marks**] A triangular field has a perimeter of $(12x^2 - 5x + 9)$ meters. Two of its side lengths are $(3x^2 + 2x - 4)$ meters and $(4x^2 - 6x + 7)$ meters.

(a) Find the algebraic expression for the length of the third side.

(b) If $x = 3$, calculate the actual numerical perimeter and the lengths of all three individual sides in meters.

Q16. [**5 Marks**] What expression should be subtracted from $(x^3 - 3x^2y + 3xy^2 - y^3)$ so that the remaining expression equals $(2x^3 - x^2y + xy^2 - 3y^3)$?

Q17. [**6 Marks**] Consider the three algebraic expressions:

$$P = 3u^2 - 4uv + v^2, \quad Q = u^2 + 2uv - 3v^2, \quad R = -2u^2 + 5uv + 2v^2$$

(a) Show step-by-step that $P + Q + R = 2u^2 + 3uv$.

(b) Evaluate the numerical value of $P + Q + R$ when $u = -1$ and $v = 2$.

(c) Calculate the value of $(P - Q + R)$ when $u = -1$ and $v = 2$.

Q18. **[6 Marks]** The total cost (in or Rs.) of purchasing x textbooks, y notebooks, and z geometry boxes is modeled by the polynomial:

$$C(x, y, z) = 150x + 45y + 120z - 15$$

(a) Identify the degree of this polynomial and state its constant term.

(b) If a school orders $x = 20$ textbooks, $y = 50$ notebooks, and $z = 15$ geometry boxes, calculate the total bill amount.

(c) If a supplier offers a flat discount represented by $(20x + 5y + 10z + 35)$, find the net cost polynomial after discount and calculate the final bill for the same order.

PART II: COMPLETE DETAILED SOLUTIONS

Teacher's Note: Each solution below presents clear algebraic steps, proper distribution of negative signs, column alignments where applicable, and final boxed answers for verification.

Solutions for Section A (MCQs)

Q1. Correct Answer: (b) 6
Detailed Explanation: The degree of a term in a multi-variable polynomial is the sum of the exponents of its variables:

- Term $5x^3y^2$: degree $= 3 + 2 = 5$.
- Term $-7x^2y^4$: degree $= 2 + 4 = 6$.
- Term $3xy^3$: degree $= 1 + 3 = 4$.
- Term -11: constant, degree $= 0$.

The degree of the polynomial is the maximum term degree, which is **6**.

Q2. Correct Answer: (b) $3b(x + y) - 5a^2$
Detailed Explanation:

- "product of $3b$ and the sum of x and y" $= 3b \times (x + y) = 3b(x + y)$.
- "Five times the square of a" $= 5 \times a^2 = 5a^2$.
- "subtracting $5a^2$ from $3b(x + y)$" yields $3b(x + y) - 5a^2$.

Q3. Correct Answer: (b) $-\frac{4}{9}pq$
Detailed Explanation: Factor out p^2qr from the term:

$$-\frac{4}{9}p^3q^2r = \left(-\frac{4}{9}pq\right) \cdot (p^2qr)$$

Hence, the coefficient of p^2qr is $-\frac{4}{9}pq$.

Q4. Correct Answer: (c) $\frac{\sqrt{5}}{2}x^3 - \frac{3}{4}x^2 + \sqrt{2}x - 1$
Detailed Explanation:

- (a) contains $\sqrt{x} = x^{1/2}$ (fractional exponent $\implies$ not a polynomial).
- (b) contains $\frac{7}{x^2} = 7x^{-2}$ (negative exponent $\implies$ not a polynomial).
- (d) contains $x^{3/2}$ (fractional exponent $\implies$ not a polynomial).
- (c) has variables with non-negative integer powers $(3, 2, 1, 0)$. Coefficients $(\frac{\sqrt{5}}{2}, \sqrt{2})$ can be irrational numbers.

Q5. Correct Answer: (b) $5x^2 - 8xy + 6y^2$
Detailed Explanation:

$$(2x^2 - 3xy + 4y^2) - (-3x^2 + 5xy - 2y^2)$$

Distribute the negative sign to all terms inside the second bracket:

$$= 2x^2 - 3xy + 4y^2 + 3x^2 - 5xy + 2y^2$$

Group like terms:

$$= (2x^2 + 3x^2) + (-3xy - 5xy) + (4y^2 + 2y^2) = \mathbf{5x^2 - 8xy + 6y^2}$$

Q6. Correct Answer: (a) 38
Detailed Explanation: Substitute $x = -2$, $y = 3$, $z = -\frac{1}{2}$:

$$4x^2y = 4(-2)^2(3) = 4(4)(3) = 48$$
$$-2xyz = -2(-2)(3)\left(-\frac{1}{2}\right) = -2 \times 3 = -6$$
$$8z^2 = 8\left(-\frac{1}{2}\right)^2 = 8\left(\frac{1}{4}\right) = 2$$

Summing them together: $48 - 6 + 2 = \mathbf{38}$.

Solutions for Section B

Q7. Solution:

(i) $E_1(x, y) = \frac{3}{5}x^2y^3 - 7x^4y + 2$:
- Exponents of variables are $2, 3, 4, 1$ (all non-negative integers). $\implies$ **It is a Polynomial**.
- Term degrees: $\frac{3}{5}x^2y^3 \to 2 + 3 = 5$; $-7x^4y \to 4 + 1 = 5$; $2 \to 0$. $\implies$ **Degree = 5**.
- Number of non-zero terms $= 3 \implies$ **Trinomial**.

(ii) $E_2(a) = 4a^2 - \frac{3}{a} + 5 = 4a^2 - 3a^{-1} + 5$:
- Contains a^{-1} with negative power -1. $\implies$ **Not a Polynomial**.

Q8. Solution:

(i) In $-15x^3y^2z$, factor out x^2y:

$$-15x^3y^2z = (-15xyz) \cdot (x^2y) \implies \text{Coefficient of } x^2y = \mathbf{-15xyz}$$

(ii) In $12a^2b^3c$, factor out $-4ab$:

$$12a^2b^3c = (-3ab^2c) \cdot (-4ab) \implies \text{Coefficient of } -4ab = \mathbf{-3ab^2c}$$

Q9. Solution: Given expression: $3x^3 - 5x^2y + 2xy^2 - 7x^3 + 8x^2y - 6xy^2 + 4x^3 - 3x^2y$
Group like terms together:

$$\begin{aligned} &= (3x^3 - 7x^3 + 4x^3) + (-5x^2y + 8x^2y - 3x^2y) + (2xy^2 - 6xy^2) \\ &= (3 - 7 + 4)x^3 + (-5 + 8 - 3)x^2y + (2 - 6)xy^2 \\ &= (0)x^3 + (0)x^2y + (-4)xy^2 \\ &= \mathbf{-4xy^2} \end{aligned}$$

Q10. Solution: Let the required polynomial be P.

$$(2a^2 - 5ab + 3b^2) + P = (a^2 + 2ab - b^2)$$

Solving for P:

$$\begin{aligned} P &= (a^2 + 2ab - b^2) - (2a^2 - 5ab + 3b^2) \\ &= a^2 + 2ab - b^2 - 2a^2 + 5ab - 3b^2 \\ &= (a^2 - 2a^2) + (2ab + 5ab) + (-b^2 - 3b^2) \\ &= \mathbf{-a^2 + 7ab - 4b^2} \end{aligned}$$

Solutions for Section C

Q11. Solution:

(i) Sum of squares $= p^2 + q^2$; Thrice product $= 3pq$. Expression: $\frac{p^2+q^2}{3pq}$.

(ii) Seven times $m = 7m$; Quotient of n divided by $5 = \frac{n}{5}$. Expression: $7m - \frac{n}{5}$.

(iii) Excess $= (5x^2 - 2xy + 3y^2) - (2x^2 + 3xy - 4y^2)$

$$= 5x^2 - 2xy + 3y^2 - 2x^2 - 3xy + 4y^2 = \mathbf{3x^2 - 5xy + 7y^2}$$

Q12. Solution:
Step 1: Calculate First Sum $S_1 = (3x^2 - 5x + 7) + (-2x^2 + 8x - 11)$:

$$S_1 = (3-2)x^2 + (-5+8)x + (7-11) = x^2 + 3x - 4$$

Step 2: Calculate Second Sum $S_2 = (5x^2 - 2x + 1) + (x^2 - 3x - 4)$:

$$S_2 = (5+1)x^2 + (-2-3)x + (1-4) = 6x^2 - 5x - 3$$

Step 3: Subtract S_1 *from* S_2:

$$S_2 - S_1 = (6x^2 - 5x - 3) - (x^2 + 3x - 4) = 6x^2 - 5x - 3 - x^2 - 3x + 4 = \mathbf{5x^2 - 8x + 1}$$

Q13. Solution:
Given $A = 2x^2 - 3xy + y^2$, $B = -x^2 + 2xy + 3y^2$, $C = 3x^2 - xy - 2y^2$:

(i) $A + B - C$:

$$\begin{aligned} A + B &= (2x^2 - x^2) + (-3xy + 2xy) + (y^2 + 3y^2) = x^2 - xy + 4y^2 \\ (A+B) - C &= (x^2 - xy + 4y^2) - (3x^2 - xy - 2y^2) \\ &= x^2 - xy + 4y^2 - 3x^2 + xy + 2y^2 = \mathbf{-2x^2 + 6y^2} \end{aligned}$$

(ii) $2A - 3B + C$:

$$\begin{aligned} 2A &= 4x^2 - 6xy + 2y^2 \\ -3B &= 3x^2 - 6xy - 9y^2 \\ C &= 3x^2 - xy - 2y^2 \\ \text{Sum} &= (4+3+3)x^2 + (-6-6-1)xy + (2-9-2)y^2 = \mathbf{10x^2 - 13xy - 9y^2} \end{aligned}$$

Q14. Solution:
Perimeter formula for rectangle: $P = 2(\text{Length} + \text{Breadth}) = 2(L + B)$.
Given $P = 14x^2 + 6x - 8$ and $L = 4x^2 - x + 3$:

$$L + B = \frac{P}{2} = \frac{14x^2 + 6x - 8}{2} = 7x^2 + 3x - 4$$

Now solve for B:

$$\begin{aligned} B &= (7x^2 + 3x - 4) - L \\ &= (7x^2 + 3x - 4) - (4x^2 - x + 3) \\ &= 7x^2 + 3x - 4 - 4x^2 + x - 3 \\ &= \mathbf{3x^2 + 4x - 7} \text{ units} \end{aligned}$$

Solutions for Section D

Q15. Solution:

(a) Let the three sides be S_1, S_2, S_3. Given $P = 12x^2-5x+9$, $S_1 = 3x^2+2x-4$, $S_2 = 4x^2-6x+7$:

$$S_1 + S_2 = (3x^2 + 4x^2) + (2x - 6x) + (-4 + 7) = 7x^2 - 4x + 3$$

$$S_3 = P - (S_1 + S_2) = (12x^2 - 5x + 9) - (7x^2 - 4x + 3)$$
$$S_3 = 12x^2 - 5x + 9 - 7x^2 + 4x - 3 = \mathbf{5x^2 - x + 6} \text{ meters}$$

(b) Evaluate for $x = 3$:

- $P = 12(3)^2 - 5(3) + 9 = 12(9) - 15 + 9 = 108 - 15 + 9 = \mathbf{102}$ m.
- $S_1 = 3(3)^2 + 2(3) - 4 = 27 + 6 - 4 = \mathbf{29}$ m.
- $S_2 = 4(3)^2 - 6(3) + 7 = 36 - 18 + 7 = \mathbf{25}$ m.
- $S_3 = 5(3)^2 - (3) + 6 = 45 - 3 + 6 = \mathbf{48}$ m.

Check: $29 + 25 + 48 = 102$ m (Verified!).

Q16. Solution:
Let the required expression to be subtracted be M.

$$(x^3 - 3x^2y + 3xy^2 - y^3) - M = (2x^3 - x^2y + xy^2 - 3y^3)$$

Solving for M:

$$\begin{aligned} M &= (x^3 - 3x^2y + 3xy^2 - y^3) - (2x^3 - x^2y + xy^2 - 3y^3) \\ &= x^3 - 3x^2y + 3xy^2 - y^3 - 2x^3 + x^2y - xy^2 + 3y^3 \\ &= (x^3 - 2x^3) + (-3x^2y + x^2y) + (3xy^2 - xy^2) + (-y^3 + 3y^3) \\ &= \mathbf{-x^3 - 2x^2y + 2xy^2 + 2y^3} \end{aligned}$$

Q17. Solution:

(a) $P + Q + R = (3u^2 - 4uv + v^2) + (u^2 + 2uv - 3v^2) + (-2u^2 + 5uv + 2v^2)$:

$$u^2\text{-terms: } 3 + 1 - 2 = 2u^2$$

$$uv\text{-terms: } -4 + 2 + 5 = 3uv$$
$$v^2\text{-terms: } 1 - 3 + 2 = 0v^2$$

Hence, $P + Q + R = \mathbf{2u^2 + 3uv}$ (Proved!).

(b) Substitute $u = -1$, $v = 2$:

$$P + Q + R = 2(-1)^2 + 3(-1)(2) = 2(1) - 6 = \mathbf{-4}$$

(c) For $P - Q + R$:

$$\begin{aligned} P - Q + R &= (3u^2 - 4uv + v^2) - (u^2 + 2uv - 3v^2) + (-2u^2 + 5uv + 2v^2) \\ &= 3u^2 - 4uv + v^2 - u^2 - 2uv + 3v^2 - 2u^2 + 5uv + 2v^2 \\ &= (3 - 1 - 2)u^2 + (-4 - 2 + 5)uv + (1 + 3 + 2)v^2 \\ &= 0u^2 - uv + 6v^2 = -uv + 6v^2 \end{aligned}$$

Substituting $u = -1, v = 2$: $-(-1)(2) + 6(2)^2 = 2 + 6(4) = 2 + 24 = \mathbf{26}$.

Q18. Solution:

(a) In $C(x, y, z) = 150x + 45y + 120z - 15$:

- Maximum power of any variable is 1 $\implies$ **Degree = 1**.
- Constant term $= -15$.

(b) Substitute $x = 20, y = 50, z = 15$:

$$C = 150(20) + 45(50) + 120(15) - 15 = 3000 + 2250 + 1800 - 15 = \mathbf{7035}$$

(c) Discount $D = 20x + 5y + 10z + 35$. Net Cost $C_{\text{net}} = C - D$:

$$\begin{aligned} C_{\text{net}} &= (150x + 45y + 120z - 15) - (20x + 5y + 10z + 35) \\ &= (150 - 20)x + (45 - 5)y + (120 - 10)z + (-15 - 35) \\ &= \mathbf{130x + 40y + 110z - 50} \end{aligned}$$

Evaluating final bill for $x = 20, y = 50, z = 15$:

$$C_{\text{net}} = 130(20) + 40(50) + 110(15) - 50 = 2600 + 2000 + 1650 - 50 = \mathbf{6200}$$